

Data Quality Documentation

FinTrust Digital Bank — Week 2 Data Analytics Track, Part A

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Companion documents: Customer_Data_Cleaning_Log and Transaction_Data_Cleaning_Log

1. Purpose and Scope

This document summarises the data quality assessment performed on FinTrust's two core datasets ahead of Week 2 analysis: the issues identified, their potential impact on downstream analysis, and the key findings from each quality check performed. It covers structural checks, missing values, duplicates, data-type issues, unusual values, statistical outliers, and the relationship between the two datasets.

2. Datasets Assessed

Dataset	Records	Original Columns	Cleaned Columns	Source
Customer_Data	1,500	12	12	FinTrust_Customer_Data.csv
Transaction_Data	12,000	11	11	FinTrust_Transaction_Data.csv

The two datasets are linked via Customer_ID, a one-to-many relationship (one customer to many transactions).

3. Assessment Methodology

Each dataset was profiled programmatically (Python/Pandas) rather than reviewed manually, given the volume involved (12,000 transaction rows). The following checks were performed on both datasets: record and column counts, missing-value detection (including hidden blank strings), full-row and primary-key duplicate detection, categorical value inventory (checking for inconsistent casing or spelling), numeric range and distribution checks, statistical outlier detection (IQR method), data-type verification, and cross-dataset referential-integrity checks.

4. Customer Data — Findings

Check	Result	Assessment
Missing values	0 across all 12 columns	Fully populated — no action required
Full-row duplicates	0	No duplicate customer records
Duplicate Customer_ID (primary key)	0	Customer_ID is a valid, unique identifier

Check	Result	Assessment
Duplicate Customer_Name	1,500 rows share a name with another row	Not a data-quality defect — a small synthetic name pool is reused; Customer_ID, not name, is the correct primary key
Categorical fields (Gender, City, Segment, Account_Type, Income Band, Preferred_Channel, Account_Status)	No inconsistent casing, spelling, or stray categories found	Clean, well-controlled categorical data
Numeric ranges (Age, Tenure_Months, Digital_Engagement_Score)	All values within plausible bounds (Age 18-65, no negative tenure, engagement 0-100 scale)	No out-of-range values

5. Transaction Data — Findings

Check	Result	Assessment
Missing values	96 rows (0.8%) missing Device_Type; 96 rows (0.8%) missing Location	Data dictionary identifies these missing values as expected characteristics of the dataset. They were retained as missing during profiling and are represented as Unknown during cleaning
Full-row / primary-key duplicates	0	No duplicate transaction records
Negative or zero Amount_NGN	0	No invalid transaction amounts
Transaction_DateTime data type	Stored as text, not a real datetime	Data-type issue — would block any date-based filtering, sorting, or trend analysis until corrected
Categorical fields (Type, Channel, Device_Type, Location, International flag, Status, Risk flag)	No inconsistent casing, spelling, or stray categories found	Clean, well-controlled categorical data
Amount_NGN distribution	Right-skewed: mean NGN46,706 vs median NGN10,306. IQR method flags 12.3% of transactions as statistical outliers	Expected for transaction-value data (many small transfers, fewer large payments) — not a data error. Reporting the mean alone without the median would misrepresent a 'typical' transaction

6. Cross-Dataset Relationship Assessment

Check	Result
Relationship type	One customer to many transactions, joined on Customer_ID

Check	Result
Orphan transactions (Customer_ID not found in Customer_Data)	No orphan transactions were identified
Customers with zero transactions	0
Transactions per customer	Minimum 1, median 8, maximum 20
Transactions linked to Dormant customer accounts	866
Transactions linked to Restricted customer accounts	224

Referential integrity between the two tables is fine as every transaction has a valid, matching customer record. The 1,090 transactions (866 + 224) linked to Dormant or Restricted accounts are a logical inconsistency worth flagging: accounts in these statuses would not normally be expected to show transaction activity.

7. Summary of Issues Identified and Business Impact

Issue	Business Impact if Left Unaddressed
Transaction_DateTime stored as text	Blocks any date-based sorting, filtering, or trend analysis (e.g. Part C's monthly transaction trend) until corrected
96 + 96 missing Device_Type / Location values	If not handled properly, missing column values could appear as blank categories and complicate grouped analysis, making channel/location reporting less complete and reducing interpretability
12.3% of transactions statistically flagged as outliers	If dropped naively, would remove legitimate high-value transactions — real business signal, not noise — from the analysis
1,090 transactions on Dormant/Restricted accounts	If not flagged, EDA and dashboard outputs could imply these account statuses are fully inactive when they are not, misrepresenting account-status-based reporting
1,500 shared Customer_Name values	A naive duplicate check based on name would incorrectly flag all customer records as duplicates, risking incorrect record removal

8. Conclusion

Both datasets are structurally sound: zero true duplicates, perfect referential integrity, and clean categorical data throughout. The issues identified are minor in scope (under 1% of transaction rows affected by missing values) and, in each case, are either confirmed as intentional (per the data dictionary) or represent expected characteristics of real-world financial data (right-skewed transaction values) rather than data corruption. Full detail on the specific action taken for each issue is recorded in the accompanying Data Cleaning Log (Excel). With these issues addressed, both datasets are assessed as ready for the SQL, Python EDA, and Power BI work that follows.